

Bahram Behzadian

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Google Scholar | U.S. permanent resident

Reinforcement Learning | Post-Training | Evaluation

Profile

RL researcher (Ph.D.) working on policy improvement with imperfect evaluators and verifier feedback, including finite-depth error bounds for recurrent value evaluators (UPI-TRM) and robust MDPs under model uncertainty. At Meta, I build correctness and monitoring systems for large-scale online experimentation over ranking models, detecting stale, biased, or inconsistent signals. Interested in RL post-training for reasoning and long-horizon agents, including durable memory and reliable tool-use and search decisions under uncertainty.

Selected Research

Unrolled Policy Iteration for Tiny Recursive Models

ICLR 2026 Workshop on AI with Recursive Self-Improvement

Co-developed an analysis of policy improvement with truncated recurrent evaluators and verifier feedback. The workshop paper separates Bellman-residual error from truncation bias under an inner-loop contraction assumption and links centered advantage error to conservative policy-improvement bounds. Follow-on work develops finite-reference error bounds without assuming recurrent convergence, separating endpoint discrepancy from reference Bellman-residual error, with an exact finite-MDP example and diagnostics on a fixed set of Sudoku states.

Robust and Risk-Sensitive Markov Decision Processes

NeurIPS 2021 and AISTATS 2021

Developed scalable algorithms for L_∞ -constrained S-rectangular robust MDPs and percentile criteria under model uncertainty.

Experience

2022–Present Research Scientist, *Meta*

- Work on a Python reference implementation of ads experiment metrics for independent cross-checks of SQL-based correctness calculations.
- Develop rule-based data-quality monitoring and statistical diagnostics for stale, biased, or inconsistent signals.
- Turn production failure analyses into automated checks for overlapping data slices and attribution double counting in online experimentation over ranking models.

2018 AI Engineer Intern, *Envio, Inc.*, Dover, NH

Developed learning-based methods for vehicle routing and scheduling under operational constraints and uncertain execution.

Selected Publications

Bahram Behzadian, Brett Daley, Gopeshh Subbaraj, Houssam Nassif
Unrolled Policy Iteration for Tiny Recursive Models
ICLR 2026 Workshop on AI with Recursive Self-Improvement [Paper]

Bahram Behzadian, Marek Petrik, Chin Pang Ho
Fast Algorithms for L_∞ -Constrained S-Rectangular Robust MDPs
Conference on Neural Information Processing Systems (NeurIPS), 2021 [Paper]

Bahram Behzadian, Reazul Hasan Russel, Marek Petrik, Chin Pang Ho
Optimizing Percentile Criterion using Robust MDPs
International Conference on Artificial Intelligence and Statistics (AISTATS), 2021 [Paper]

Bahram Behzadian, Soheil Gharatappeh, Marek Petrik
Fast Feature Selection for Linear Value Function Approximation
International Conference on Automated Planning and Scheduling (ICAPS), 2019 [Paper]

Bahram Behzadian, Pratik Agarwal, Wolfram Burgard, Gian Diego Tipaldi
Monte Carlo Localization in Hand-Drawn Maps
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2015 [Paper]

Complete publication list: [Publications page](#) | [Google Scholar](#)

Technical Skills

Programming: Python, PyTorch.

Evaluation and experimentation: experiment-metric correctness, statistical diagnostics, anomaly and data-quality monitoring, ML auditing.

RL and post-training: robust MDPs, policy improvement, verifier-style rewards and RL with verifiable rewards; hands-on PPO and DPO experiments on small language models for arithmetic; working knowledge of GRPO and reward modeling.

Education

- 2015–2022 Ph.D. in Computer Science, *University of New Hampshire*
Dissertation: *Efficient Data-Driven Robust Policies for Reinforcement Learning* (advisor: Marek Petrik).
M.S. in Computer Science awarded en route (2018). Thesis: *Representation Learning for Reinforcement Learning via Singular Value Decomposition*.
- 2010–2013 M.S. in Machine Automation, *Tampere University*
Thesis: *Robot Localization under Partial Observability with Weak Maps*

Honors and Fellowships

CEPS Graduate Fellowship, University of New Hampshire
Thesis and Dissertation Fellowship, Tampere University of Technology